

EE 14 Final project proposal

Overview

For your final project, you will work with a team of 3 to build something awesome with what you've learned from this course.

The first step is to submit a 1-2 page proposal of what you plan to create, so we have clear plan of what you're working on, and so the teaching staff can give you advice to make your project a success. Read on for the details!

At the highest level, the only requirement for your project is that it should take input from one or more devices, do "something interesting" on the STM32, and produce one or more kind of output. Practically, this means you can build basically anything you can imagine with a microcontroller in it, from a wrist-mounted Pac-Man game to an alarm clock that pours water on your head.

Past teams have built things like the following — but now that we've switched to the Nucleo board and opened up the possibilities with the ADC and I2C, we think you can raise the bar even higher!

- Infrared TV remote (e.g., TV-B-Gone)
- 2-wheel mobile robot
- Smartwatch / pedometer
- Karaoke machine / portable audio player
- Audio synthesizer
- Party games like Heads Up, Catch Phrase, Bop It
- Tabletop flick football goalpost
- Plant babysitter

This project will be the most fun if you go beyond the parts we've worked with. Your team should implement at least two "new things" — either new hardware peripherals on the STM32 (e.g., the DAC), or a new device that interfaces with your STM32 using GPIO, the ADC, I2C, SPI, or UART. Essentially anything you can find on Digikey, Sparkfun, or Adafruit is fair game.

We're willing to spend about \$30 per team; if you want to buy more expensive stuff (GPS modules or LIDAR or wall-sized displays) you're welcome to do so on your own dime.

Additionally, we have some of the following things on hand:

- Materials from labs so far (buttons, LEDs, 7-segment displays, joysticks, IMUs, temperature sensors)
- HX 11 ultrasonic distance sensors
- Sharp IR distance sensors
- PS/2 computer keyboards
- NES controllers from ES 4
- Wii Nunchucks
- SD cards (for storing lots of extra data)
- Stuff in the Nolop store including servos, DC motors and motor drivers (you'll need to purchase these yourself)

Project proposal

Your project proposal should be a brief document (1-2 pages) which explains the main idea of the project and clearly defines what it will do. It should contain the following:

1. **Project description:** One or two paragraphs describing your project: the basic idea, what the input device(s) will be, what the output device(s) will be, and how people will interact with it.
2. **Physical drawing:** This should be one or more sketches of the physical design of the system, with appropriate annotations to explain how users will interact with it.

3. **Block diagram:** This should be a diagram showing the main hardware components and how they will connect. You should show individual wires and where they will connect (e.g., which power supply you're using for each component, and which things connect to which GPIO pins).
4. **Hardware needed:** A list of any hardware components you will need for the project beyond the items we already have (listed above). In addition to listing this in your proposal document, please add these to the spreadsheet with the name of a member of your team.

Submit your proposal on Gradescope; please submit one per team (you should be able to do a group submission on Gradescope) since that makes it much easier for us to give feedback.